

Total no. of Pages: 2

Roll no.

Degree: B.Tech.(ECAM), Semester: 3rd — Course work
END-SEMESTER EXAMINATION (Odd), Nov-Dec 2025

Course Code: EAEPC-304
Duration: 3 Hours

Course Title: Microelectronics Circuits & Applications
Max. Marks: 50

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Questions	Marks	CO
Q.1	Attempt any two parts of the following		
1a	Classify and explain amplifiers based on their frequency response with the help of neat magnitude response curves of each of them.	5	CO1
1b	Derive the condition for small signal operation for a CS amplifier given in Fig. 1b below. Derive the transconductance (g_m) expression in terms of overdrive voltage (V_{ov}) and the expression for voltage gain (A_v).	5	CO1
1c	In the Fig. 1c below, find the effective resistance between drain and source, when the drain voltage is 0.1V. Let $V_t=1V$ and $K'_n (W/L)=1mA/V^2$.	5	CO1
Q.2	Attempt any two parts of the following		
2a	In the circuit of Fig. 2a, determine the voltages at all nodes and the currents through all branches. Assume $\beta = 100$.	5	CO2
2b	Fig. 2b below shows a CS amplifier utilizing the drain-to-gate feedback biasing arrangement. The input signal v_i is coupled to the gate via a large capacitor, and the output signal at the drain is coupled to the load resistance R_L via another large capacitor. Determine its small-signal voltage gain, its input resistance, and the largest allowable input signal. The transistor has $V_t = 1.5 V$, $k'_n (W/L) = 0.25 mA/V^2$, and $V_A = 50 V$.	5	CO2
2c	Explain the MOS differential pair operation with a common-mode input voltage and with a differential input voltage.	5	CO2
Q.3	Attempt any two parts of the following		
3a	Derive the common-mode gain (A_{cm}) expression for the difference operational amplifier given in Fig. 3a.	5	CO3
3b	Derive closed-loop gain expressions for inverting and non-inverting configurations of operational amplifier.	5	CO3
3c	Derive the transfer function expression for the circuit given in Fig. 3c.	5	CO3
Q.4	Attempt any two parts of the following		
4a	Explain the working of a monostable multivibrator using the 555 IC.	5	CO4
4b	Explain the principle of working and draw the transfer characteristics of the Op-Amp based bi-stable multivibrator circuit.	5	CO4
4c	Explain Phase-locked loop (PLL). What are the various applications of PLL?	5	CO4
Q.5	Attempt any two parts of the following		
5a	What are the basic principles of sinusoidal oscillators? Explain	5	CO5

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	Barkhausen criterion of oscillation.		
5b	Explain any three important properties of negative feedback in amplifier circuits.	5	CO5
5c	Explain the characteristics of Power MOSFETs. Also compare Power MOSFETs with Power BJTs.	5	CO5

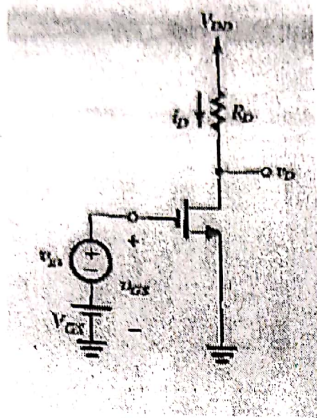


Fig: 1b

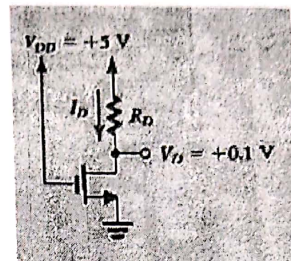


Fig: 1c

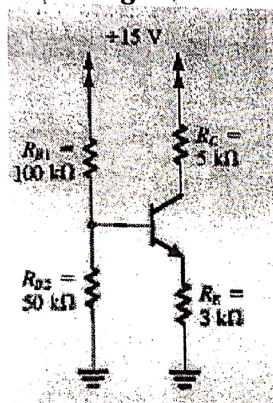


Fig: 2a

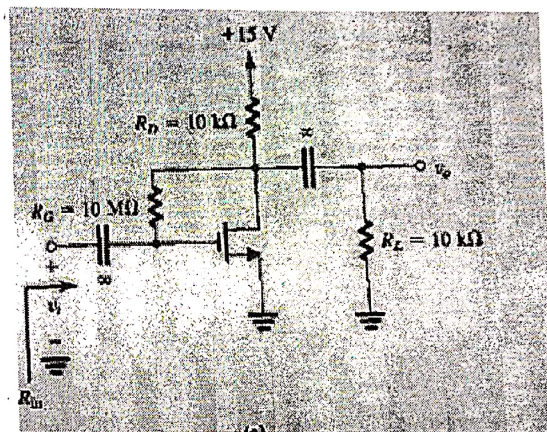


Fig: 2b

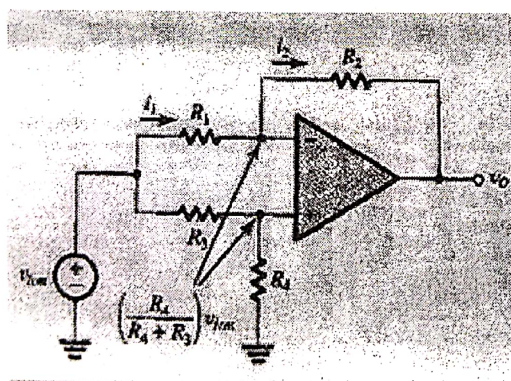


Fig: 3a

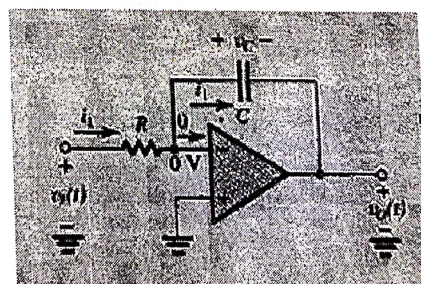


Fig: 3c

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